Usage-based AI exposure across Australia's earnings distribution

Summary of a working draft

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Descriptive; not a job-loss or career-risk score; not advice.

The data behind this work are US task-level Claude.ai use by self-selected users, from one product. No Australian occupational AI use is measured. The scores describe tasks in US occupations; this work applies them to the occupations Australians work in, and compares them with measures of what AI could do that do not come from Anthropic.

The question

How is measured AI use distributed across Australia's earnings distribution? And where does measured use run ahead of, or behind, what AI could do? Australian monitoring of generative AI has so far relied on measures of potential, such as Jobs and Skills Australia's exposure scores. Knowing which earners' occupations AI is actually used in comes before any question about wages or jobs.

What was done

Anthropic's usage-based "observed exposure" scores (Massenkoff and McCrory, 2026) were linked from US occupations to the 358 ANZSCO unit groups used in ABS Labour Force data, through the US and international occupation classifications. Each unit's score is an employment-weighted average of the US occupations linked to it. Units were then ranked by ABS weekly earnings and grouped into ten employment-weighted deciles, and each unit's rank on usage was compared with its rank on Jobs and Skills Australia's automation and augmentation scores and on Eloundou et al.'s measure of what language models could do.

The scores cover 99.1% of Australian employment. Every result is repeated under four ways of linking the classifications and on the units with the cleanest links. Results are reported whether or not they hold.

What the figures show

All figures are for the main specification. The alternatives are in the note's appendix.

- **Usage against potential.** Measured use ranks the lowest-earning fifth of employment higher than two of the three potential measures do (JSA augmentation and Eloundou et al.) and the highest-earning fifth lower. Against JSA's automation scores the pattern runs the other way, and it is not robust at the bottom end across ways of linking the classifications.
- **Higher earners.** Usage-based exposure averages 0.07 in the bottom earnings decile and 0.11 in the top, though not in a straight line. Most of the difference is whether an occupation's tasks show any measured use at all: that share rises from 0.39 to 0.81. How heavily use shows up, once it does, has no clear pattern.
- **Concentration.** Usage-based exposure is concentrated among higher earners (concentration index 0.09): less than Eloundou et al.'s measure, more than either JSA measure.
- **Disagreements.** The largest disagreements with JSA's automation scores are mostly clerical, driving and hospitality-management occupations that measured use ranks well below JSA.
- **Women and men.** Women's average usage-based exposure (0.14) is above men's (0.10) in every earnings decile. This reflects the occupations they work in.
- **Western Australia.** WA's average (0.107) is below the national one (0.120), entirely because of occupation mix, mostly its smaller share of professionals.

What this does not show

The usage data come from one product's users, and the source classifies the task in each conversation, not the job of the person having it. Over half of the scored US occupations score zero, which means their tasks fell below a usage threshold or failed the capability gate, not that no one uses AI for them. The comparison with Eloundou et al.'s measure is partly mechanical, because the usage-based score already requires a task to be feasible on that measure. Nothing here estimates whether AI changes employment, earnings or hours.

Read more

- Working draft, appendix and occupation explorer: https://129313531343-bit.github.io/ai-wage-distribution/
- Code, data sources and decision log: https://github.com/129313531343-bit/ai-wage-distribution

Written with Claude Code, an AI coding assistant, which wrote the analysis and drafted this summary under the author's direction; the author reviewed and approved it. See the repository's "Contributions and AI use" section.